Replication Report

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Introduction

For my replication exercise, I decided to replicate Schoonvelde et al.'s 2019 paper: "*Liberals lecture, conservatives communicate: Analyzing complexity and ideology in 381,609 political speeches*". In this paper, the authors ask whether the well-documented finding that liberal politicians use more complex language than conservatives, holds across countries, time periods and political contexts. To test this question, they analyze 380 000 speeches from three datasets: ParlSpeech (5 European parliaments), EUSpeech (12 prime minister) and party congresses from Denmark and the Netherlands. Once they have the speeches, the authors measured complexity using the Flesch-Kincaid grade score. In addition to complexity, they also measured speaker ideology using the Manifesto Project's ideology dataset. Once they had their data cleaned, the authors ran a standard OLS, regressing the Flesch-Kincaid score on ideology, time and whether the politician was currently in power. The authors find that culturally liberal politicians tend to use more complex language than their conservative counterparts, that speeches are becoming less complex over time, and that the economic dimension of ideology doesn't have any effect.

Now that I have gone over the paper, I will explain my main goal for this replication project. My main goal is to replicate the results as thoroughly as possible, using raw, uncleaned, data. To do this, I will be using the original datasets (downloaded from their original sources) and attempt to recreate every step that the authors describe in their paper. Rather than relying on the author's pre-cleaned data, I will be starting from scratch.

Results and Differences

To save space and for the sake of simplicity, I will be discussing the results and the differences in the same section. I will begin by explaining the results that the original authors created, then I will show my results and compare. Please refer to the appendix to see the figures.

We are going to begin with the descriptive statistics. This is a great place to start, as it shows us descriptive information about our cleaned datasets, and it will let us know if there are any major discrepancies before we run the main analyses. Comparing our tables, you can see that my datasets have consistently fewer observations than the original data. Some of my datasets have a relatively low difference (such as n = -119), while others, like the British parliamentary data, have a larger difference (n = - 37 076). This is slightly concerning but is likely due to differences in filtering choices and ideological placement (which we will discuss further in the replication autopsy). Looking at the means for our variables, we can see that they mostly align. Our variables for complexity, "progressive – conservative" (cultural ideology), year and "in-government" are all well aligned. The "left-right" measure (economic ideology) are consistently different than the original data, and it seems to be skewed to the right (the max and mins seem to be slightly larger). This isn't too much of a concern as the author's are unable to find relationships with this variable anyways. Overall, the descriptive statistics look acceptable, and we are good to continue with our analyses.

Figure 1 shows some descriptive information about the linguistic complexity of selected speakers. This shows that the Flesch-Kincaid grade scores work. In the author's figure, they select 8 speakers. For my replication, not all the 8 speakers selected remained in my final dataset. For this reason, I only showed the scores of 4 speakers. Though the numbers of speakers are different,

my figure replicates the same ordering as the original, showing that the scores are working as the original authors intended.

Figure 2 shows the main results of the study, the OLS regressions of complexity on ideology. The authors show that liberals consistently have more complex speeches than conservatives (the cultural dimension). Though the results are less conclusive for the economic dimension of ideology. My results replicate this, apart from the United Kingdom, which is likely due to differences in cleaning.

Figure 3 shows the effect OLS regression of complexity by time. The author's results show that, for all the datasets except for the United Kingdom, speech complexity is decreasing over time. My results replicate this, except for the United Kingdom. In my results, complexity in the U.K. was increasing but started declining in the 2010s. I thought that I had made a mistake in my replication, but Professor Vallejo mentioned that when he ran similar analyses on similar data, he got similar results. So, I'm inclined to believe that perhaps my analysis is correct and the author's is questionable. This connects to the assumptions I had to make during the cleaning, which we will talk about shortly.

Figure 4 shows the effects of ideology on complexity, over time. It shows that, with fixed effects for party, parties tend to become less complex as they become more conservative. This figure in particular showcases the effect of ideology on complexity over time very well.

Lastly, the author also provides the summary tables for their OLS results. Overall, my replication matches these results, though we will not spend time examining them as the figures show the important points.

Replication Autopsy

Now that we have gone through the results of the original study as well as my replication efforts, let's walk through what it took to get here. Beginning with obtaining data, the authors only offered their pre-cleaned data for their analyses. For this reason, I chose to replicate the entire procedure, as I wasn't comfortable trusting their validation packages. So instead of using what they had, I decided to download the original data from their original sources, and to follow their procedure as identically as possible to replicate their results. Unfortunately, the authors didn't mention each exact detail, and thus I had to make some assumptions. This is the main discrepancy between the different number of observations between our datasets, and likely the reason why some of our results differ slightly. But all that to say that the first step of this replication was downloading the raw data. I found these datasets on Harvard's Dataverse (speech data) as well as their personal websites (Manifesto Project). Once I had the data, I removed unimportant variables that took up space and allowed myself to create a subset of the sample via sampling. This was important for testing reasons, as the full dataset took a very long time to process, so I didn't want to run the full dataset each time.

Once I had the data, I was ready to clean it. This was mostly unproblematic. I followed the instructions that the authors gave regarding each dataset and calculated the ideology scores using the formulas provided. One issue was that the EUSpeech dataset didn't contain party information, so I had to use the internet to add the relevant information based on their names. Once I had all the party information, as well as the manifesto project, I was able to add ideological information to the datasets based on their parties. For this, it was important to keep track of the years as well, as the parties' ideologies change by election. Lastly, I had to make sure that all the party names between the datasets and the manifesto project lined up.

Next, I had to calculate the complexity scores. This is where I realized I was using the wrong EUSpeech dataset (a pre-cleaned, punctuation removed dataset) and had to switch to the right one. I also had to split the ParlSpeech dataset into 3 chunks so that my computer wouldn't run out of memory. I also calculated standardized scores, as the authors did. Once I got that set up, I was able to quickly add whether or not the speaker was a member of the governing party at the time. Finally, I was done cleaning and prepared a final dataset with standardized variable names.

Once I had the clean dataset, the analyses were very straight forward. I used simple functions and packages to measure the descriptive statistics and make the tables and figures. Once that was complete, I just had to interpret my results and compare with the original findings.

Extension

There would be lots of interesting ways to extend this paper. One way would be in using a different complexity measure. The Flesch-Kincaid grade scores seem to work well for English speeches but extending it beyond English is questionable. For that reason, it would be interesting to use a different measure of complexity to test across countries, as well as different countries that aren't in the anglosphere. I would also like to see these methods applied to new data. Both ParlSpeech and EUSpeech have had newer releases, and it would be interesting to see if the findings hold. It would also be interesting to expand it past WEIRD countries, and to examine speech complexity outside of Europe and north America. Lastly, using the same data, you could ask new research questions as well. For example, you could see if specific shocks to countries have effects on speech complexity (such as financial collapse or terrorist attacks). Alternatively, it would be interesting to directly test how personalities match onto speech complexity, as the authors explain the effect of conservatism on speech though that mechanism.

References:

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Appendix

Descriptive Statistics (Tables A1-A8)

Author's Data

Table A.1: Party congresses DNK

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	528	−0.046	0.771	−1.396	−0.513	0.315	6.435
Left-Right	528	−0.222	1.624	−3.925	−1.156	1.312	3.104
Progressive-Conservative	528	0.405	1.688	−2.938	−0.672	1.120	6.037
Year	528	44.670	15.520	0	33.8	57	69
In government	528	0.347	0.476	0	0	1	1

Table A.2: Party congresses NLD

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	659	−0.057	0.697	−1.493	−0.583	0.418	2.486
Left-Right	659	−0.080	0.889	−2.011	−0.721	0.633	1.807
Progressive-Conservative	659	−0.582	0.905	−2.543	−1.140	−0.133	2.622
Year	659	39.420	20.514	1	22	58	72
In government	659	0.584	0.493	0	0	1	1

Table A.3: Parliament GER

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	60,763	0.150	0.488	−2.030	−0.192	0.451	6.756
Left-Right	60,763	−0.326	1.185	−2.921	−1.235	0.740	1.485
Progressive-Conservative	60,763	−0.290	0.972	−2.912	−0.889	0.189	1.336
Year	60,763	12.342	6.714	0	7	19	22
In government	60,763	0.548	0.498	0	0	1	1

Table A.4: Parliament ESP

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	35,154	2.502	1.051	−1.795	1.786	3.143	9.028
Left-Right	35,154	−0.213	0.790	−2.239	−0.859	0.495	1.234
Progressive-Conservative	35,154	−0.853	0.662	−2.576	−1.397	−0.300	0.206
Year	35,154	13.212	7.264	0	7	20	26
In government	35,154	0.371	0.483	0	0	1	1

Table A.5: Parliament GBR

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	161,439	−0.178	0.465	−2.716	−0.461	0.109	8.640
Left-Right	161,439	0.146	0.705	−0.841	−0.448	0.684	1.666
Progressive-Conservative	161,439	0.527	0.745	−1.124	−0.028	0.839	1.844
Year	161,439	14.014	7.718	0	7	21	27
In government	161,439	0.545	0.498	0	0	1	1

Table A.6: Parliament SWE

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	72,955	−0.728	0.414	−3.088	−1.006	−0.472	2.093
Left-Right	72,955	0.152	1.364	−3.260	−0.831	0.891	3.613
Progressive-Conservative	72,955	−0.501	0.998	−5.112	−0.995	0.254	2.056
Year	72,955	13.051	7.294	0	7	19	25
In government	72,955	0.391	0.488	0	0	1	1

Table A.7: Parliament NLD

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	48,281	−0.383	0.378	−2.109	−0.629	−0.175	4.625
Left-Right	48,281	−0.140	0.929	−3.100	−0.672	0.403	1.469
Progressive-Conservative	48,281	0.006	0.991	−2.613	−0.739	0.593	2.622
Year	48,281	11.664	6.098	0	6	17	21
In government	48,281	0.551	0.497	0	0	1	1

Table A.8: Heads of Government

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	1,830	0.089	0.756	−2.210	−0.446	0.547	5.865
Left-Right	1,830	0.010	0.760	−2.356	−0.921	0.684	1.102
Progressive-Conservative	1,830	−0.079	1.005	−3.628	−0.487	0.828	3.110

My Data

Table A.1: Party congresses DNK

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	451	-0.045	1.081	-1.793	-0.654	0.379	11.297
Left-Right	451	0.926	1.714	-2.915	-0.393	1.950	4.263
Progressive-Conservative	451	0.302	1.348	-2.148	-0.565	0.837	4.710
Year	451	45.550	15.494	0.000	35.000	58.000	69.000
In government	451	0.118	0.322	0.000	0.000	0.000	1.000

n = -71

Table A.2: Party congresses NLD

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	540	0.038	0.926	-1.913	-0.621	0.653	3.269
Left-Right	540	0.872	0.866	-1.784	0.246	1.455	3.120
Progressive-Conservative	540	-0.515	0.641	-1.867	-0.978	-0.110	2.446
Year	540	37.350	21.076	0.000	19.750	57.000	71.000
In government	540	0.219	0.414	0.000	0.000	0.000	1.000

n = -119

Table A.3: Parliament GER

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	61,512	0.120	0.488	-1.866	-0.220	0.423	4.651
Left-Right	61,512	1.056	1.123	-0.629	0.152	2.034	3.391
Progressive-Conservative	61,512	-0.313	0.822	-2.435	-0.786	0.117	1.194
Year	61,512	12.104	6.711	0.000	6.000	18.000	22.000
In government	61,512	0.515	0.500	0.000	0.000	1.000	1.000

n = +749

Table A.4: Parliament ESP

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	34,622	2.077	1.004	-1.062	1.377	2.701	8.214
Left-Right	34,622	1.099	0.789	-0.400	0.515	1.659	2.898
Progressive-Conservative	34,622	-0.699	0.551	-2.637	-1.182	-0.281	0.162
Year	34,622	13.660	7.313	0.000	8.000	20.000	26.000
In government	34,622	0.346	0.476	0.000	0.000	1.000	1.000

n = -532

Table A.5: Parliament GBR

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	124,363	-0.197	0.737	-2.369	-0.640	0.163	78.973
Left-Right	124,363	0.805	0.725	-1.121	0.214	1.464	1.751
Progressive-Conservative	124,363	0.367	0.612	-1.547	0.000	0.780	2.998
Year	124,363	14.226	7.655	0.000	8.000	21.000	27.000
In government	124,363	0.498	0.500	0.000	0.000	1.000	1.000

n = -37 076

Table A.6: Parliament SWE

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	65,535	-0.573	0.509	-2.452	-0.930	-0.246	2.146
Left-Right	65,535	0.428	1.371	-2.160	-0.388	1.387	3.660
Progressive-Conservative	65,535	-0.545	0.909	-4.117	-1.103	0.197	1.587
Year	65,535	13.270	7.258	0.000	7.000	19.000	25.000
In government	65,535	0.382	0.486	0.000	0.000	1.000	1.000

n = -7 420

Table A.7: Parliament NLD

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	44,258	-0.389	0.544	-2.186	-0.742	-0.091	6.266
Left-Right	44,258	0.963	0.789	-0.350	0.364	1.357	3.382
Progressive-Conservative	44,258	0.103	0.818	-2.043	-0.399	0.594	2.446
Year	44,258	11.692	6.100	0.000	6.000	17.000	21.000
In government	44,258	0.486	0.500	0.000	0.000	1.000	1.000

n = -4023

Table A.8: Heads of Government

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	1,442	0.000	1.000	-2.886	-0.718	0.650	4.341
Left-Right	1,442	0.919	0.705	-0.473	0.214	1.659	2.584
Progressive-Conservative	1,442	-0.054	0.836	-2.059	-0.387	0.780	2.312
Year	1,442	4.394	2.469	0.000	2.000	7.000	8.000
In government	1,442	0.917	0.276	0.000	1.000	1.000	1.000

n = -388

Figures (1-4)

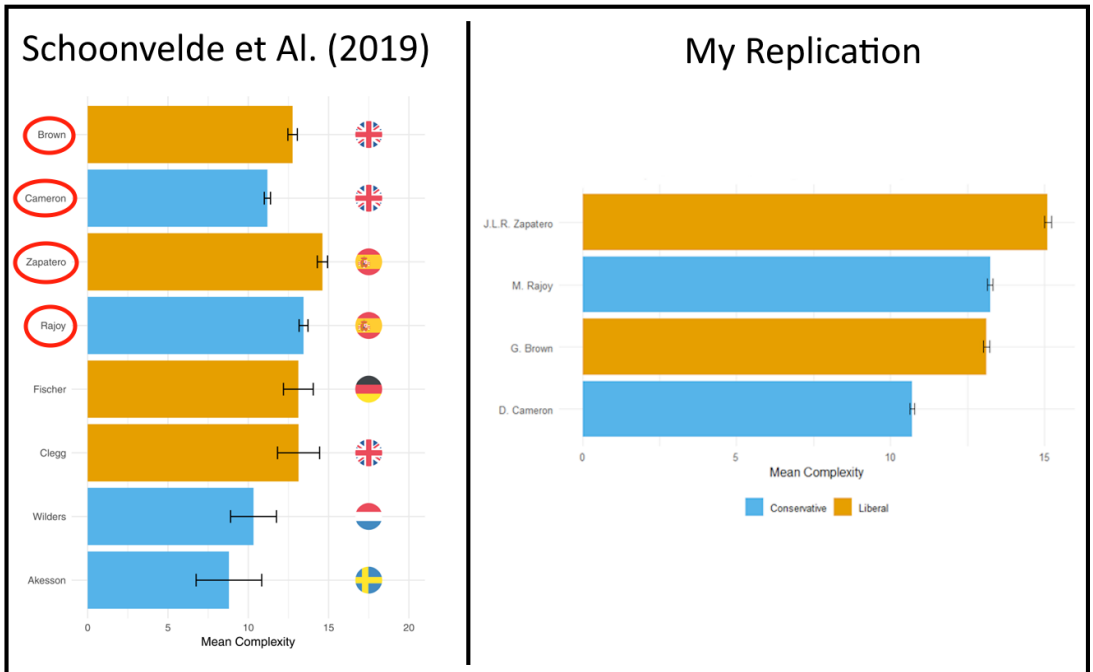

Figure 1: Descriptive Information on Linguistic Complexity

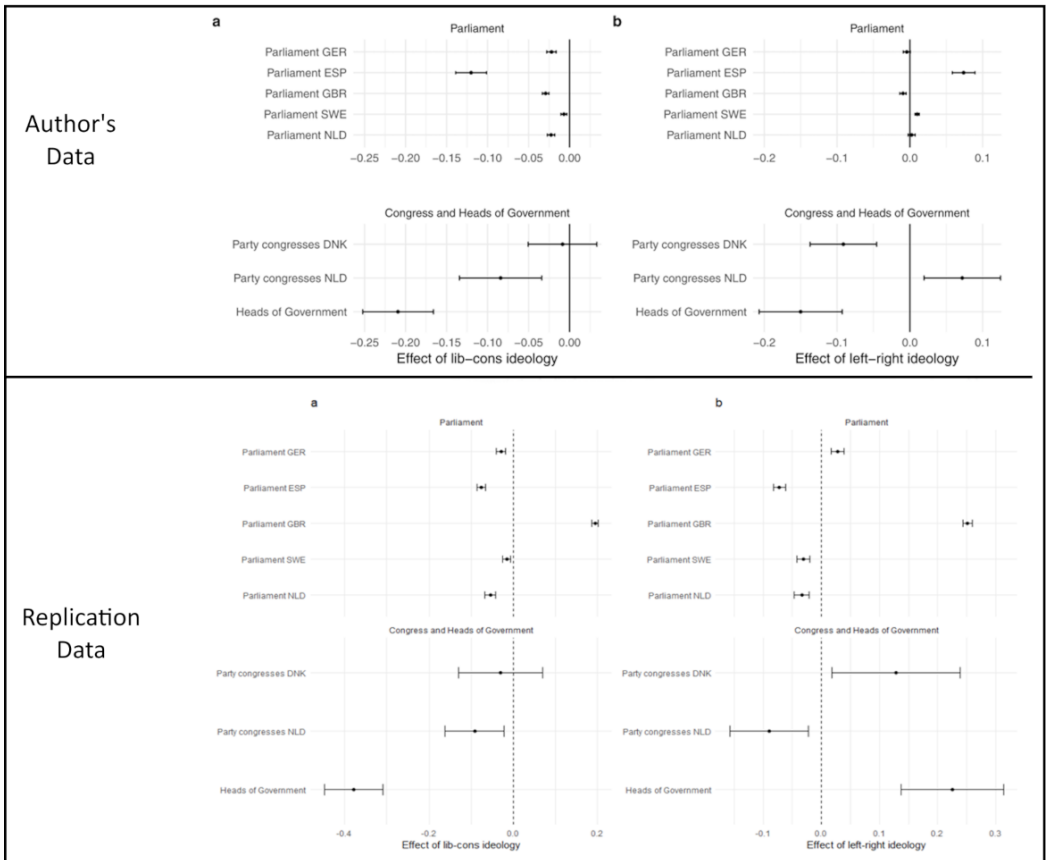

Figure 2: OLS Regression on Ideology

Author's Data

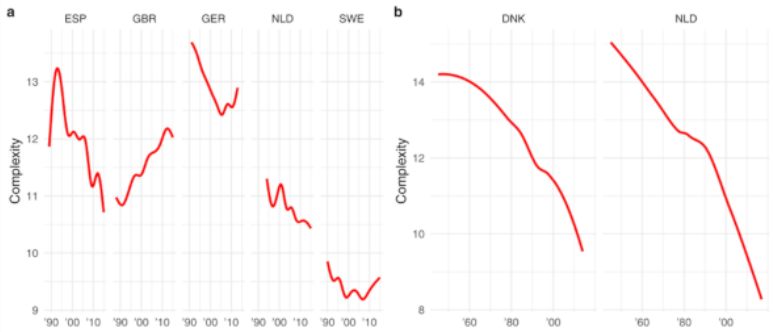

Fig 3. OLS regression of complexity on time. Plots *a* an *b* display loess regression lines of average speech complexity over time in the parliamentary speeches and congress speeches respectively. These are local estimates of the effect of year on complexity.

https://doi.org/10.1371/journal.pone.0208450.g003

Replication Data

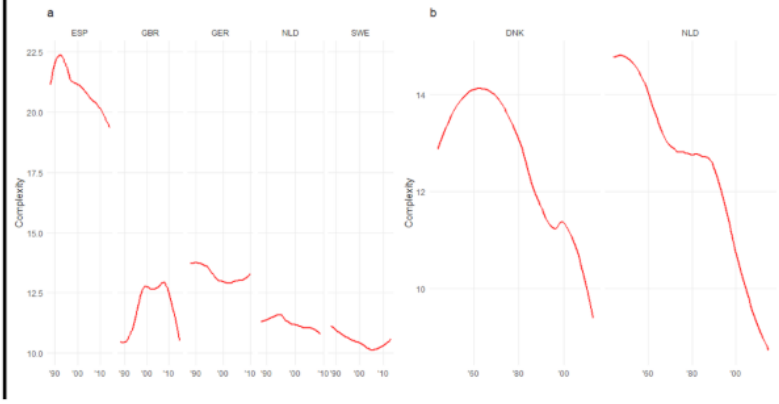

Figure 3: OLS Regression on Complexity on Time

Author's Data

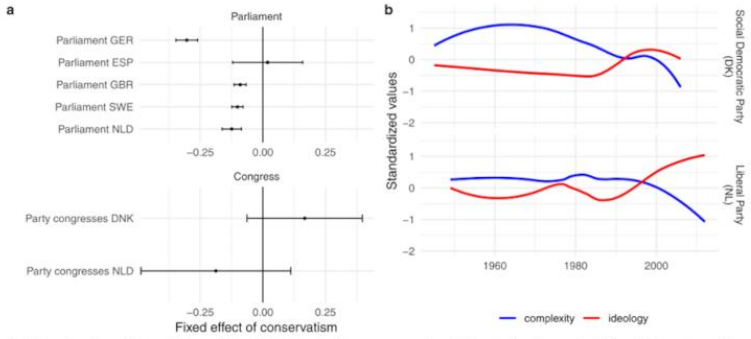

Fig 4. Regression of complexity on ideology with fixed effects for party. Plot *a* reports regression coefficients for liberal conservative ideology in 7 regression models with fixed effects for party (one for each dataset in the corpus with the exception of the prime minister speeches). Plot *b* displays loess regression lines of average speech complexity over time for the Danish social democratic party and the Dutch liberal party, as well as their ideological position.

https://doi.org/10.1371/journal.pone.0208450.g004

Replication Data

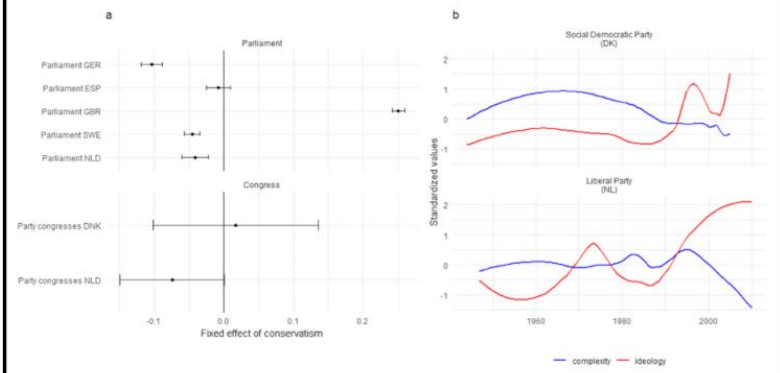

Figure 4: OLS Regression on Complexity of Time

OLS Results

Party Fixed Effects (Table C1)

Author's Results

Table C.1: OLS regression results

	DNK Congress	NLD Congress	GER Parl	ESP Parl
Left-Right	−0.091***	0.072***	−0.004*	0.074***
	(0.023)	(0.027)	(0.002)	(0.008)
Progressive-Conservative	−0.009	−0.084***	−0.022***	−0.120***
	(0.021)	(0.026)	(0.003)	(0.010)
Year	−0.021***	−0.022***	−0.009***	−0.020***
	(0.002)	(0.001)	(0.0003)	(0.001)
In government	0.159**	0.075*	0.039***	−0.020*
	(0.066)	(0.044)	(0.004)	(0.012)
Constant	0.836***	0.736***	0.235***	2.682***
	(0.113)	(0.061)	(0.005)	(0.015)
Observations	528	659	60,763	35,154
R^2	0.181	0.481	0.022	0.026
Adjusted R^2	0.175	0.478	0.022	0.026
Residual Std. Error	0.700 (df = 523)	0.504 (df = 654)	0.483 (df = 60758)	1.037 (df = 35149)
F Statistic	28.868*** (df = 4; 523)	151.337*** (df = 4; 654)	348.687*** (df = 4; 60758)	235.073*** (df = 4; 35149)
Note:				*p<0.1; **p<0.05; ***p<0.01

	GBR Parl	SWE Parl	NLD Parl	PM
Left-Right	−0.010***	0.010***	0.002	−0.150***
	(0.002)	(0.001)	(0.002)	(0.029)
Progressive-Conservative	−0.029***	−0.007***	−0.023***	−0.209***
	(0.002)	(0.002)	(0.002)	(0.022)
Year	0.014***	−0.001**	−0.007***	
	(0.0001)	(0.0003)	(0.0003)	
In government	−0.058***	0.071***	0.038***	
	(0.002)	(0.003)	(0.004)	
Constant	−0.328***	−0.753***	−0.326***	0.074***
	(0.003)	(0.005)	(0.005)	(0.016)
Observations	161,439	72,955	48,281	1,830
R^2	0.065	0.009	0.020	0.157
Adjusted R^2	0.065	0.009	0.020	0.156
Residual Std. Error	0.450 (df = 161434)	0.412 (df = 72950)	0.375 (df = 48276)	0.695 (df = 1827)
F Statistic	2,809.898*** (df = 4; 161434)	164.229*** (df = 4; 72950)	246.653*** (df = 4; 48276)	169.938*** (df = 2; 1827)
Note:				*p<0.1; **p<0.05; ***p<0.01

Replication Results

Table C.1: OLS regression results

	Dependent variable: flesch_kincaid			
	DNK Congress (1)	NLD Congress (2)	GER Parl (3)	ESP Parl (4)
Left-Right	0.081**	-0.096***	0.012***	-0.092***
	(0.035)	(0.037)	(0.003)	(0.007)
Progressive-Conservative	-0.024	-0.123***	-0.017***	-0.139***
	(0.041)	(0.051)	(0.003)	(0.010)
Year	-0.031***	-0.029***	-0.007***	-0.028***
	(0.004)	(0.002)	(0.0003)	(0.001)
In government	0.085	0.088	0.034***	-0.052***
	(0.150)	(0.085)	(0.004)	(0.011)
Constant	1.267***	1.106***	0.174***	2.478***
	(0.157)	(0.072)	(0.006)	(0.015)
Observations	451	540	61,512	34,622
R^2	0.162	0.454	0.016	0.054
Adjusted R^2	0.154	0.447	0.016	0.054
Residual Std. Error	0.994 (df = 446)	0.689 (df = 535)	0.484 (df = 61507)	0.976 (df = 34617)
F Statistic	21.520*** (df = 4; 446)	109.809*** (df = 4; 535)	246.720*** (df = 4; 61507)	497.892*** (df = 4; 34617)
Note:				*p<0.1; **p<0.05; ***p<0.01

Table C.1 continued

	Dependent variable: flesch_kincaid			
	GBR Parl (1)	SWE Parl (2)	NLD Parl (3)	PM (4)
Left-Right	0.055***	-0.011***	-0.023***	0.052***
	(0.004)	(0.002)	(0.004)	(0.048)
Progressive-Conservative	0.233***	-0.009***	-0.036***	-0.442***
	(0.004)	(0.003)	(0.004)	(0.041)
Year	0.006***	-0.006***	-0.008***	
	(0.0003)	(0.0004)	(0.0004)	
In government	-0.024***	0.087***	-0.032***	
	(0.004)	(0.004)	(0.006)	
Constant	-0.563***	-0.532***	-0.258***	-0.255***
	(0.006)	(0.006)	(0.008)	(0.048)
Observations	124,363	65,535	44,258	1,442
R^2	0.054	0.018	0.011	0.267
Adjusted R^2	0.054	0.018	0.011	0.266
Residual Std. Error	0.717 (df = 124358)	0.504 (df = 65530)	0.541 (df = 44253)	0.857 (df = 1439)
F Statistic	1,765.100*** (df = 4; 124358)	306.168*** (df = 4; 65530)	122.660*** (df = 4; 44253)	261.871*** (df = 2; 1439)

Table C.1: Party Fixed Effects

Speaker Fixed Effects (Table D1)

Author's Results

Table C.1: OLS regression results

	DNK Congress	NLD Congress	GER Parl	ESP Parl
Left-Right	−0.091***	0.072***	−0.004*	0.074***
	(0.023)	(0.027)	(0.002)	(0.008)
Progressive-Conservative	−0.009	−0.084***	−0.022***	−0.120***
	(0.021)	(0.026)	(0.003)	(0.010)
Year	−0.021***	−0.022***	−0.009***	−0.020***
	(0.002)	(0.001)	(0.0003)	(0.001)
In government	0.159**	0.075*	0.039***	−0.020*
	(0.066)	(0.044)	(0.004)	(0.012)
Constant	0.836***	0.736***	0.235***	2.682***
	(0.113)	(0.061)	(0.005)	(0.015)
Observations	528	659	60,763	35,154
R^2	0.181	0.481	0.022	0.026
Adjusted R^2	0.175	0.478	0.022	0.026
Residual Std. Error	0.700 (df = 523)	0.504 (df = 654)	0.483 (df = 60758)	1.037 (df = 35149)
F Statistic	28.868*** (df = 4; 523)	151.337*** (df = 4; 654)	348.687*** (df = 4; 60758)	235.073*** (df = 4; 35149)
Note:				*p<0.1; **p<0.05; ***p<0.01

	GBR Parl	SWE Parl	NLD Parl	PM
Left-Right	−0.010***	0.010***	0.002	−0.150***
	(0.002)	(0.001)	(0.002)	(0.029)
Progressive-Conservative	−0.029***	−0.007***	−0.023***	−0.209***
	(0.002)	(0.002)	(0.002)	(0.022)
Year	0.014***	−0.001**	−0.007***	
	(0.0001)	(0.0003)	(0.0003)	
In government	−0.058***	0.071***	0.038***	
	(0.002)	(0.003)	(0.004)	
Constant	−0.328***	−0.753***	−0.326***	0.074***
	(0.003)	(0.005)	(0.005)	(0.016)
Observations	161,439	72,955	48,281	1,830
R^2	0.065	0.009	0.020	0.157
Adjusted R^2	0.065	0.009	0.020	0.156
Residual Std. Error	0.450 (df = 161434)	0.412 (df = 72950)	0.375 (df = 48276)	0.695 (df = 1827)
F Statistic	2,809.898*** (df = 4; 161434)	164.229*** (df = 4; 72950)	246.653*** (df = 4; 48276)	169.938*** (df = 2; 1827)
Note:				*p<0.1; **p<0.05; ***p<0.01

Replication Results

Table C.1: OLS regression results

	Dependent variable: flesch_kincaid			
	DNK Congress (1)	NLD Congress (2)	GER Parl (3)	ESP Parl (4)
Left-Right	0.081**	-0.096***	0.012***	-0.092***
	(0.035)	(0.037)	(0.003)	(0.007)
Progressive-Conservative	-0.024	-0.123***	-0.017***	-0.139***
	(0.041)	(0.051)	(0.003)	(0.010)
Year	-0.031***	-0.029***	-0.007***	-0.028***
	(0.004)	(0.002)	(0.0003)	(0.001)
In government	0.085	0.088	0.034***	-0.052***
	(0.150)	(0.085)	(0.004)	(0.011)
Constant	1.267***	1.106***	0.174***	2.478***
	(0.157)	(0.072)	(0.006)	(0.015)
Observations	451	540	61,512	34,622
R^2	0.162	0.454	0.016	0.054
Adjusted R^2	0.154	0.447	0.016	0.054
Residual Std. Error	0.994 (df = 446)	0.689 (df = 535)	0.484 (df = 61507)	0.976 (df = 34617)
F Statistic	21.520*** (df = 4; 446)	109.809*** (df = 4; 535)	246.720*** (df = 4; 61507)	497.892*** (df = 4; 34617)
Note:				*p<0.1; **p<0.05; ***p<0.01

Table C.1 continued

	Dependent variable: flesch_kincaid			
	GBR Parl (1)	SWE Parl (2)	NLD Parl (3)	PM (4)
Left-Right	0.055***	-0.011***	-0.023***	0.052***
	(0.004)	(0.002)	(0.004)	(0.048)
Progressive-Conservative	0.233***	-0.009***	-0.036***	-0.442***
	(0.004)	(0.003)	(0.004)	(0.041)
Year	0.006***	-0.006***	-0.008***	
	(0.0003)	(0.0004)	(0.0004)	
In government	-0.024***	0.087***	-0.032***	
	(0.004)	(0.004)	(0.006)	
Constant	-0.563***	-0.532***	-0.258***	-0.255***
	(0.006)	(0.006)	(0.008)	(0.048)
Observations	124,363	65,535	44,258	1,442
R^2	0.054	0.018	0.011	0.267
Adjusted R^2	0.054	0.018	0.011	0.266
Residual Std. Error	0.717 (df = 124358)	0.504 (df = 65530)	0.541 (df = 44253)	0.857 (df = 1439)
F Statistic	1,765.100*** (df = 4; 124358)	306.168*** (df = 4; 65530)	122.660*** (df = 4; 44253)	261.871*** (df = 2; 1439)

Table D.1: Party Fixed Effects